Building Segmentation of Hazy Aerial Images via Collaborative Decoupling Knowledge Learning

Shan Wang, Libao Zhang, *Member, IEEE*, and Weisi Lin, *Fellow, IEEE*

***Abstract*—Building segmentation from aerial images is critical for applications such as urban planning and disaster management. However, haze severely degrades image quality, leading to radiance distortion, reduced contrast, and inaccurate segmentation. These degradations introduce two major challenges: 1) conventional methods typically treat dehazing and segmentation as separate tasks, which causes error accumulation and limits performance in haze-laden scenarios; and 2) the diverse scales of buildings and haze-induced domain shifts hinder robust feature representation and accurate boundary delineation. To address these issues, we propose a collaborative decoupling knowledge learning (CDKL) framework that jointly performs dehazing and segmentation within a unified model. Specifically, a dual-branch decoupled network (D-Net) separates image restoration and segmentation, while a bidirectional feature co-interaction module (BFCIM) enables cross-task guidance: segmentation features refine dehazing for clearer structural details, and dehazing features enhance segmentation to resolve haze-induced ambiguities. Furthermore, a teacher network (T-Net) pretrained on clear aerial images guides D-Net through a collaborative knowledge transfer (CKT) system, which aligns features across domains using multilevel strategies, thereby mitigating distribution shifts and improving robustness. Extensive experiments on synthetic and real hazy aerial datasets demonstrate that CDKL achieves state-of-the-art performance, delivering significant gains in segmentation accuracy and robustness across diverse haze conditions. The source code is available at https://github.com/Shan-rs/CDKL.git**

***Index Terms*—Building segmentation, dehazing, feature transfer, knowledge distillation, remote sensing.**

I. Introduction

BUILDING segmentation of aerial images is crucial for numerous applications, including land-use analysis, urban planning, and environmental monitoring [1]. However, haze often significantly degrades the quality of aerial images, leading to challenges in extracting accurate semantic information. The presence of haze causes severe radiance distortion and reduces image contrast, complicating object identification and segmentation. Traditional segmentation methods for aerial images are commonly developed under the assumption that the input images are clear and free of any atmospheric interference [2], [3]. This assumption simplifies the segmentation process, as clear images allow segmentation algorithms to detect object boundaries and textures accurately. However, in real-world applications, aerial images are frequently degraded by various atmospheric conditions such as haze, cloud cover, motion blur, and sensor noise. These atmospheric effects cause significant deterioration in image quality, often obscuring important visual details and making it challenging to distinguish between objects and regions within an image. Haze, in particular, arises from suspended particles in the atmosphere, and this scattering and attenuation of light further reduces the contrast of the image, making object identification difficult [4].

Statistics show that around 35% of the Earth's land surface remains consistently covered by clouds and haze throughout the year [5]. As a consequence, haze is one of the most prevalent sources of degradation in aerial images, and this presents a substantial obstacle to accurate semantic segmentation. Segmenting objects in such degraded aerial images becomes particularly challenging when relying on methods optimized solely for clear images.

Most existing segmentation algorithms, which are optimized for clear aerial images, tend to perform poorly when applied directly to hazy aerial images. As shown in Fig. 1(c), semantic segmentation models trained exclusively on clear aerial images often fail to accurately delineate objects and regions of interest in hazy conditions. This poor performance can be attributed to several challenges introduced by hazy atmospheric conditions.

1) *Distribution Shift:* The presence of haze alters the feature distribution within hazy aerial images, causing a mismatch with the feature distribution of clear aerial images. The result is a "distribution shift," making it difficult for models trained on clear aerial images to generalize effectively to hazy conditions [6], [7]. Without accounting for this shift, models often misinterpret haze as part of the object itself or mistake object boundaries, leading to substantial errors in segmentation.
2) *Complex Target Size Variability and Weak Signals:* Aerial images often contain objects of diverse sizes, including numerous small and densely packed buildings, which exacerbates the problem of insufficient target information. These small targets are more susceptible to interference from haze, resulting in blurred

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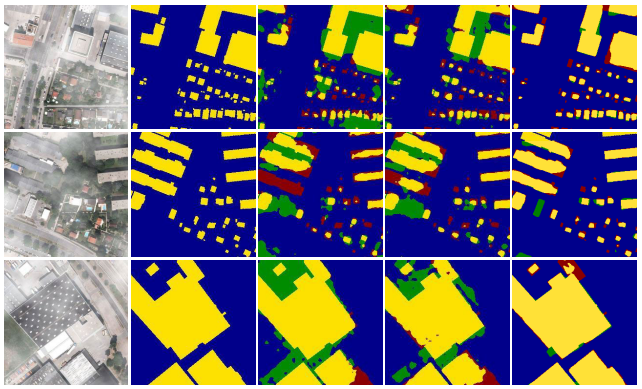


Fig. 1. Examples of building segmentation results of hazy aerial images. Different colors are used for a better view, that is, yellow for true positives, blue for true negatives, green for false positives, and red for false negatives. (a) Hazy aerial images. (b) Ground truths. (c) DeepLabv3+ [13]. (d) DCIL [12] + DeepLabv3+ [13]. (e) CDKL (ours).

or indistinguishable boundaries. Accurately identifying and segmenting these objects requires the model to be sensitive to small features, which is difficult in hazy conditions [8], [9].

3) *Complex Scene Structure and Dense Object Overlap:* Unlike natural images, where foreground objects tend to be prominent and well-separated from the background, aerial images contain complex scenes with closely intertwined foreground and background elements [10], [11]. This intricate spatial arrangement in aerial images causes foreground and background elements to overlap, which makes it challenging to separate them accurately. Consequently, aerial image segmentation models depend heavily on high-quality input data and struggle to adapt to noisy samples such as hazy images, resulting in errors in object boundary detection and misclassification of regions.

Addressing these challenges necessitates the development of robust models capable of handling diverse atmospheric interferences (e.g., haze and sensor noise) while maintaining accuracy across varying object scales. This requires architectural innovations that integrate restoration techniques such as dehazing directly with segmentation algorithms to mitigate degradation effects. Furthermore, such models must be adaptable to domain shifts, ensuring resilience in complex and variable remote sensing environments.

A widely adopted approach to tackle the segmentation challenges posed by hazy aerial images involves a two-step process: first, applying dehazing techniques to enhance the aerial images, followed by feeding the enhanced images into a segmentation model. For instance, as shown in Fig. 1(d), dehazing the aerial images with dynamic collaborative inference learning (DCIL) [12] before applying a segmentation model like DeepLabv3+ [13] improves segmentation results over directly segmenting hazy aerial images, as shown in Fig. 1(c). However, while this enhancement-then-segmentation approach can improve segmentation performance to some degree, it overlooks the inherent connection between the processes of dehazing and segmentation. The result is a system with complex processing steps, limited robustness, and suboptimal segmentation performance, as the dehazing and segmentation processes are treated independently, rather than being optimized in a unified manner.

To address the above challenges, we propose a collaborative decoupling knowledge learning (CDKL) model specifically designed for building a segmentation of hazy aerial images. Unlike conventional two-stage approaches, CDKL integrates dehazing and segmentation within a unified framework, providing targeted solutions to the three major challenges.

First, to alleviate the distribution shift between clear and hazy aerial images, CDKL incorporates a teacher network (T-Net) pretrained on clear images. Through the collaborative knowledge transfer (CKT) system, T-Net provides robust feature guidance and transfers knowledge to the dual-branch network (D-Net), ensuring consistent feature representations across domains. The CKT system achieves this alignment via multilevel strategies, including Gram matrix-based feature alignment, logit mimicking, and pseudo-pixel regression, thereby reducing the mismatch in feature distributions.

Second, to handle complex target size variability and weak signals, CDKL introduces a decoupled D-Net that separates the tasks of dehazing and segmentation. Within the segmentation branch, a feature interaction module (FIM) dynamically adjusts the receptive field (RF) through deformable convolutions to capture building structures of different scales. This enables the network to remain sensitive to small and weak targets, ensuring more accurate recognition and segmentation under hazy conditions.

Third, to overcome complex scene structures and dense object overlaps, CDKL integrates a bidirectional feature co-interaction module (BFCIM) into D-Net. This module enables mutual refinement between the dehazing and segmentation branches: dehazing features provide structural cues to sharpen object boundaries, while segmentation features offer semantic priors to guide restoration. This bidirectional collaboration significantly improves fine-detail restoration and robust boundary localization in complex aerial scenes.

Through these targeted designs, CDKL effectively addresses distributional, structural, and semantic challenges in hazy aerial image segmentation, ensuring both robustness and accuracy under diverse atmospheric conditions. In summary, the main contributions of this work are as follows.

1) *CDKL Model*: We propose a novel CDKL model to integrate both dehazing and segmentation within a single framework, improving segmentation accuracy in hazy aerial images by treating dehazing and segmentation as interrelated tasks.
2) *Decoupled Network Based on Dual Branches (D-Net)*: We design D-Net to decouple input features into haze-related and semantic features, optimizing the model's ability to handle hazy aerial images effectively and efficiently, thereby enabling robust segmentation.
3) *BFCIM*: We design the BFCIM to address haze-induced feature distortion and semantic ambiguity. By dynamically fusing dehazing features and segmentation

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semantics, it enhances fine-detail restoration and object boundary localization under low-visibility conditions.

4) *CKT System*: We introduce a new CKT system that conducts multistage feature alignment to bridge the feature gap between clear and hazy aerial images, significantly boosting the performance of the proposed training scheme.

II. Related Work

A. Low-Level Aerial Image Dehazing

According to the different principles of dehazing, aerial image dehazing methods can be divided into prior- and learning-based methods. Summarizing these methods is essential for our work, as the proposed CDKL framework explicitly incorporates a dehazing branch whose design is informed by these advances. Understanding their strengths and limitations clarifies the motivation for integrating dehazing and segmentation within a unified framework.

Prior-based methods [14], [15], [16], [17], [18], [19] are typically built upon the atmospheric scattering model and rely on hand-crafted priors. For example, Bi et al. [17] proposed a low-rank and sparse prior (LSP) method, Han et al. [18] introduced an edge-preserving multiscale decomposition approach, and Singh et al. [19] designed a fusion-based visibility restoration model. However, these methods perform well only in scenes that meet their underlying assumptions.

Learning-based dehazing has gained increasing attention in recent years. Early studies [20] estimated transmission maps under supervision, but such approaches require reference transmission maps that are rarely available. Later works adopted end-to-end networks to directly predict dehazed outputs [21], [22]. Chen et al. [23] introduced H2RL-Net to restore fine details, while Nie et al. [24] proposed contrastive haze-aware learning to better model haze distributions. GAN-based methods [25], [26] further improved generalization by training on unpaired hazy datasets. More recently, Wen et al. [27] developed RSHazeNet with enhanced skip connections, and Chi et al. [28] integrated prior- and learning-based strategies in a trinity model.

In addition, weakly supervised and semi-supervised dehazing have emerged as promising directions. Wang et al. [29] introduced a weakly supervised approach via physics-based decomposition, while another study [30] employed multiprior guidance and domain transfer for semi-supervised training. By contrast, our CDKL adopts a fully supervised paradigm with pixel-level building annotations, coupling dehazing and segmentation in a single architecture. Full supervision allows CDKL to learn precise boundaries and task-aligned features, which are further reinforced by the CKT and bidirectional co-interaction modules. These choices make CDKL particularly effective when reliable annotations are available, while remaining complementary to weakly/semi-supervised lines of work.

While the above approaches have achieved notable progress in dehazing, most focus on low-level visual quality and overlook the needs of downstream interpretation. Recent work in RSI understanding under degradation highlights the value of joint modeling. For example, Xu et al. [31] proposed CloudSeg, a multimodal framework that jointly removes clouds and performs land-cover classification with knowledge distillation. Similarly, our CDKL targets haze-degraded building segmentation, reinforcing the motivation to integrate degradation removal and semantic interpretation within a unified framework to mitigate domain shifts and retain task-relevant features.

B. High-Level Aerial Image Segmentation

Aerial image semantic segmentation at a high level has garnered significant attention in recent years. We can categorize the approaches into traditional machine learning-based methods and deep learning-based methods based on their primary methodological distinctions.

The traditional methods are typically based on classical machine learning techniques, such as support vector machines, decision trees, and random forests, to segment specific targets in aerial images. However, they can only extract some primary features, like spectral, texture-based features, which cannot meet the task requirements of semantic segmentation in complex backgrounds.

Regarding deep learning-based methods, various approaches [32], [33], [34], [35] have been proposed to address the challenges posed by the complex nature of aerial images. For example, Zhao et al. [36] proposed an attention-based segmentation network allowing the model to learn to assign different importance weights to different parts of the input images. Jin et al. [35] proposed a multimodal fusion network based on edge detection to address the semantic segmentation of multisource information fusion. Xiao et al. [37] exploited the advantages of convolution operations and the transformer to enhance multiscale representation learning. Zhou et al. [38] leveraged graph attention mechanisms and knowledge distillation techniques to enhance semantic segmentation accuracy in aerial images.

However, the above segmentation algorithms, which are optimized for clear aerial images, tend to perform poorly when applied directly to hazy aerial images. This is primarily because hazy images suffer from reduced contrast, color distortions, and blurred details, which hinder the extraction of accurate spatial and spectral features. As a result, segmentation models trained on clear images struggle to differentiate between objects or regions in hazy conditions, often misclassifying pixels or failing to capture fine details. Addressing these challenges requires specialized techniques that can either enhance the visibility of hazy aerial images before segmentation or adapt the segmentation model to be robust under various atmospheric conditions, particularly in high-haze environments.

C. Low- and High-Level Task Interaction

Recently, some researchers have attempted to use multitask learning to tackle low- and high-level task interaction for degraded aerial images within a single framework. Wang et al. [39] noticed the problem of low target recognition rates in low-resolution aerial images and proposed a target detection on

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super-resolution reconstruction (TDoSR) model. Pereira and dos Santos [40] used a joint training strategy to combine a super-resolution (SR) module with a semantic segmentation module to improve the semantic segmentation accuracy of low-resolution aerial images. Zhang et al. [41] proposed a collaborative network framework that simultaneously performs SR reconstruction and semantic segmentation of low-resolution aerial images in a shared feature extractor. Wei et al. [42] proposed a SGINet model that can sufficiently combine the interaction between the typically separate tasks of single image deraining and semantic segmentation to boost performance in both tasks simultaneously.

However, the above methods can only address the challenge of low-resolution aerial images, which fail to focus on hazy aerial images. Additionally, they do not consider the inherent feature transfer between low-quality aerial images and clear aerial images, which makes it challenging to apply them to hazy image segmentation tasks. To solve this problem, we propose the CDKL by leveraging both low-level image features and high-level semantic information, offering a robust solution for processing complex aerial images with varying levels of haze. The dual improvement underscores the framework's capacity to bridge the gap between image restoration and downstream tasks, ensuring a more holistic approach to dehazing and segmentation.

D. Knowledge Distillation

Knowledge distillation has emerged as a prominent technique for model compression, in which a student model learns the implicit knowledge from a teacher model to improve its performance. The concept was first introduced by Hinton [43], where a teacher model imparts knowledge to a student model through the soft targets of its output layer. This process leverages the softened probability distributions produced by the teacher model to encode more information about class relationships, which are not captured by traditional hard labels.

Several studies have successfully extended knowledge distillation to semantic segmentation tasks. Liu et al. [44] proposed a distillation framework for semantic segmentation, which adapted the original knowledge distillation framework [43] to focus on dense prediction tasks. By distilling both logits and intermediate feature maps, the student model learns to mimic the teacher's behavior, which is particularly useful in high-resolution images like those found in remote sensing.

In the specific domain of aerial images, Zhou et al. [38] introduced a knowledge distillation framework targeting semantic segmentation tasks of aerial images, which distills knowledge from a pretrained large T-Net and builds reliable weak labels to optimize the untrained student network. Dong et al. [45] proposed a cross-model knowledge distillation framework, distilling segmenters from CNNs and Transformers.

In summary, knowledge distillation for semantic segmentation of aerial images has seen significant progress, leveraging multiscale feature extraction, region-based focus, and attention mechanisms to improve student models' performance. These approaches address the unique challenges posed by remote sensing data and hold promise for further advancements.

The proposed method addresses an important gap in remote sensing by developing a robust segmentation method that can handle degraded atmospheric conditions. By simultaneously performing dehazing and segmentation within a unified framework, our CDKL model sets a new standard for handling hazy aerial images, with implications for other domains facing similar challenges.

III. Methodology

Our goals are twofold: 1) to promote segmentation accuracy of hazy aerial images under a unified framework and 2) to establish a more concise solution space using feature transfer. Fig. 2 illustrates the detailed structure of the proposed CDKL. We design a D-Net to achieve the first goal. And we tailor a CKT system along with knowledge distillation to achieve the second goal.

The CDKL framework is an innovative approach designed to address the challenges of building segmentation in hazy aerial images by decoupling and collaboratively optimizing the tasks of dehazing and segmentation. Its core mechanism revolves around the interaction between two networks: T-Net and D-Net, mediated by a CKT system.

A. T-Net Based on Multiscale Features

The overall structure of T-Net is illustrated in Fig. 3. As a teacher network, T-Net is designed to be more complex than the student network, providing richer feature representations to guide and optimize the latter. Specifically, we adopt four convolutional feature extraction layers from ResNet101 as the encoder to capture multiscale features. To further strengthen this ability, an atrous spatial pyramid pooling (ASPP) module [46] is integrated, which leverages dilated convolutions with multiple dilation rates to capture multiscale contextual information and thereby improve segmentation performance.

The decoder is composed of three SBlock-i ($i = 1, 2, 3$). We use the frequency–spatial hybrid gated attention (FSHGA) and residual dense block (RDB) [47] as each SBlock, which can fully utilize multilevel features and effectively restore spatial information.

1) Frequency–Spatial Hybrid Gated Attention: Prior studies [48], [49], [50] have extensively explored frequency-domain modeling and self-attention for vision tasks. However, many existing designs either rely purely on frequency modeling, which may weaken local texture details, or focus only on spatial self-attention, which lacks global context. In contrast, our FSHGA module is specifically designed to jointly exploit the complementary advantages of frequency-domain global context and spatial-domain local refinement.

As shown in Fig. 4, each input feature map is first partitioned into nonoverlapping windows, within which a 2-D fast Fourier transform (FFT) converts spatial representations into frequency spectra. Multihead self-attention is then applied in this frequency space to capture long-range dependencies across spectral coefficients. The enhanced representations are then transformed back into the spatial domain via inverse FFT. In parallel, a spatial residual convolutional branch refines local textures and boundary details. Finally, a learnable gating

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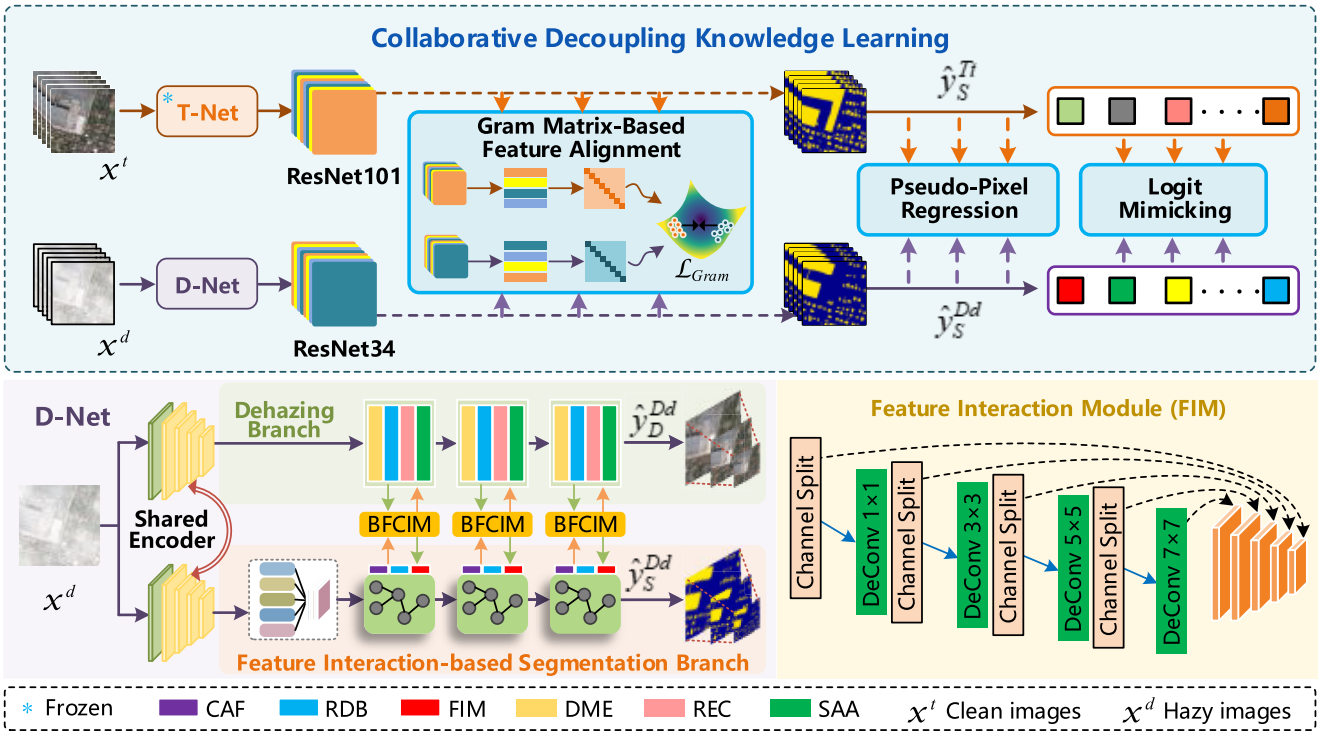


Fig. 2. Framework of the proposed CDKL. The training process of the proposed method involves D-Net and T-Net. In the inference phase, only the well-trained D-Net will be used for forward propagation to generate segmentation results.

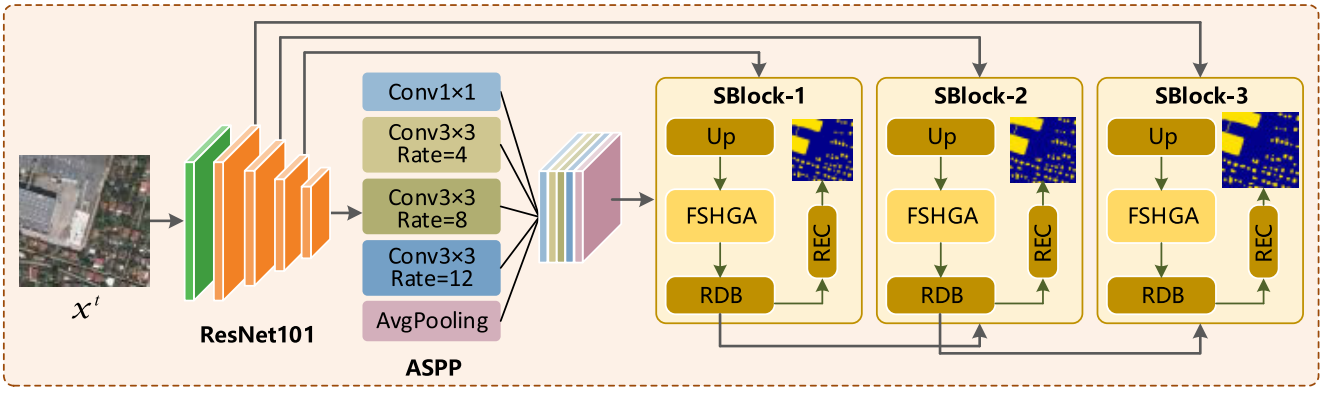


Fig. 3. Framework of T-Net.

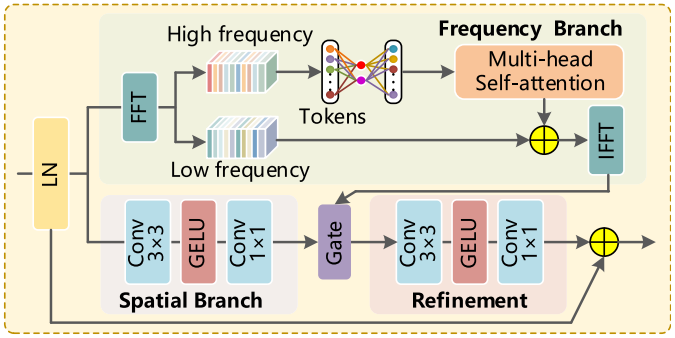


Fig. 4. Structure of FSHGA.

mechanism adaptively fuses the frequency and spatial outputs, ensuring that task-relevant information is selectively emphasized while redundant patterns are suppressed.

This hybrid attention design enables the network to simultaneously strengthen global structural awareness and fine-grained localization, which is particularly beneficial for building segmentation in clear aerial imagery where both large-scale context and precise boundaries are critical.

In T-Net, the model is optimized using T-Net loss $\mathcal{L}_T$. The $\mathcal{L}_T$ is obtained by calculating the cross-entropy loss function between the three segmentation result maps predicted by T-Net and the semantic segmentation labels

$$\mathcal{L}_T = \sum_{d=1}^{3} \lambda_d \mathcal{L}_{CE}\left(\hat{y}_S^{Td}, y_{gt}\right) \tag{1}$$

where y_{gt} represents the semantic segmentation label. λ_1, λ_2, and λ_3 represent the corresponding weight factors, which

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are set to 1/4, 1/2, and 1. The calculation formula of the cross-entropy loss function $\mathcal{L}_{CE}$ is

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_{gt}^{i}\cdot\log\hat{y}^{i} + \left(1-y_{gt}^{i}\right)\cdot\log\left(1-\hat{y}^{i}\right)\right] \tag{2}$$

where N is the total number of pixels in the image and $\hat{y}^i$ and y_{gt}^i represent the prediction and label corresponding to the ith pixel, respectively.

B. D-Net Based on Dual Branches

The D-Net consists of three parts: a shared encoder, a dehazing branch, and a segmentation branch. The detailed structure of D-Net is shown in Fig. 2.

1) Shared Encoder: The shared encoder in D-Net is built upon ResNet-34. It is designed to be similar to the encoder in T-Net, which facilitates effective feature transfer between the two networks. Unlike the T-Net encoder, however, the D-Net encoder adopts fewer convolutional layers and parameters, thereby reducing model complexity and improving segmentation efficiency during inference.

2) Dynamic Mutual Decoder-Based Dehazing Branch: The function of the dehazing branch is to decode the multilevel features in the shared encoder into specific haze-related features, thereby effectively restoring the image quality of hazy aerial images. Inspired by the dynamic mutual decoder introduced in [12], we cascade the dynamic mutual enhancement module, RDB, reconstruction module, and spectrum-aware aggregation module together to form a DBlock in the dehazing branch. The DBlock-i ($i = 1, 2, 3$) can fully extract useful information by fusing multilevel features, and effectively restore texture details and spectral features in an image by aggregating restoration feature maps of different scales.

The dehazing branch can generate three dehazing results $\hat{y}_D^{Dd}(d = 1, 2, 3)$ with different scales, which can be optimized by

$$\mathcal{L}_{Dehaze} = \sum_{d=1}^{3}\left[L_1\left(\hat{y}_D^{Dd}, G_{gt}^{d}\right) + L_{SSIM}\left(\hat{y}_D^{Dd}, G_{gt}^{d}\right)\right] \tag{3}$$

where $G_{gt}^{d}(d = 1, 2, 3)$ is ground truth. L_1 is L1-norm. L_{SSIM} denotes the loss function of structural similarity (SSIM).

3) Feature Interaction-Based Segmentation Branch: In D-Net, the purpose of the segmentation branch is to transform multiscale features from the shared encoder into specific segmentation feature maps. The detailed structure of the segmentation branch is shown in Fig. 2. An ASPP module is used to process the output of the shared encoder. By changing the dilated ratios in ASPP, the model's ability to recognize objects is enhanced.

To effectively fuse features from three levels of the shared encoder, the ASPP output, and two feature maps from the semantic segmentation branch, we design a context-aware fusion (CAF) module. As shown in Fig. 5, the process begins by sequentially feeding the ASPP output and the segmentation branch features as "Input A" into a 1×1 convolution to align the channel dimensions with "Input B."

Next, channel attention (CA) and pixel attention (PA) [51] are applied to the channel-adjusted feature maps, enhancing features both from a channel perspective and a pixel perspective. Then, the feature maps are upsampled to match the dimensions of "Input B" and passed through a global average pooling (GAP) layer followed by a Sigmoid function, yielding the weight coefficients for "Input A." Similarly, "Input B" undergoes the same process to compute its corresponding weight coefficients. Finally, the fused feature maps are calculated as follows:

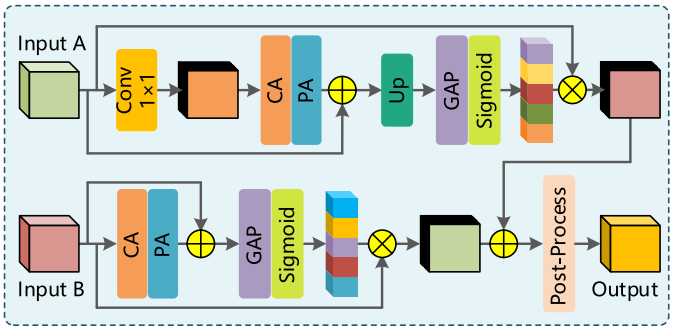


Fig. 5. Structure of the CAF module.

$$\mathcal{F}^{i} = \begin{cases} \mathcal{P}\left(\omega_1\cdot f_D^{ASPP} + \omega_2\cdot f_{SE}^{3-(i-1)}\right), & i\in\{1\} \\ \mathcal{P}\left(\omega_1\cdot f_{Seg}^{i-1} + \omega_2\cdot f_{SE}^{3-(i-1)}\right), & i\in\{2,3\} \end{cases} \tag{4}$$

where $\mathcal{P}(\cdot)$ represents the postprocessing operation, which consists of two convolutional layers with a kernel size of 3×3. $\mathcal{F}^i(i = 1, 2, 3)$ is the fused feature maps.

The fused feature maps are input into the RDB module to extract deep coding features. Then, an FIM is designed to further assist the network in focusing on the salient regions in the feature maps. The specific structure of the FIM will be detailed in Section III-C. In addition, we introduce a BFCIM to strengthen the interaction between dehazing and segmentation branches, enabling mutual guidance for clearer restoration and more accurate segmentation. Finally, a reconstruction module [12] is utilized to reconstruct the feature maps, generating segmentation result maps at three different scales. An upsampling operation is applied to upsample the three result maps to the original image size, yielding the final segmentation result maps. We show the pseudo-code implementation of the feature interaction-based segmentation branch in Algorithm 1.

By computing the cross-entropy loss between the segmentation result maps $\hat{y}_S^{Dd}(d = 1, 2, 3)$ and the semantic segmentation ground truths G_{gt}^{s}, supervision is provided for the semantic segmentation branch

$$\mathcal{L}_{seg} = \sum_{d=1}^{3}\delta_d\mathcal{L}_{CE}\left(\hat{y}_S^{Dd}, G_{gt}^{s}\right) \tag{5}$$

where δ_1, δ_2, and δ_3 are weight factors which are set to 1/4, 1/2, and 1, respectively.

C. Feature Interaction Module

The receptive field (RF) of neurons in the human visual system underpins the visual attention mechanism. A classical receptive field (CRF) consists of an excitatory center and an

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Algorithm 1 Feature Interaction-Based Segmentation Branch

Input: Feature maps $f_{SE}^{i}(i \in \{1,2,3,4\})$ in the shared encoder.

Output: The segmentation results $\hat{y}_{S}^{Di}(i = 1,2,3)$.

Compute the ASPP feature maps f_{D}^{ASPP} by
$f_{D}^{ASPP} \leftarrow \mathcal{F}_{ASPP}(f_{SE}^{4})$
for $i = 1,2,3$ **do**
  if $i == 1$ **then**
    $\mathcal{F}^{i} \leftarrow \mathcal{F}_{CAF}([f_{SE}^{3-(i-1)}, f_{D}^{ASPP}])$
  else
    $\mathcal{F}^{i} \leftarrow \mathcal{F}_{CAF}([f_{SE}^{3-(i-1)}, f_{Seg}^{i-1}])$
  end if
  Compute the high-level semantic feature maps f_{D}^{i} using RDB;
  Compute the interaction feature maps f_{Seg}^{i} using FIM;
  Refine f_{Seg}^{i} with the guidance of dehazing features by BFCIM;
  Generate the ith segmentation result $\hat{y}_{S}^{Di}$ by a reconstruction module.
end for

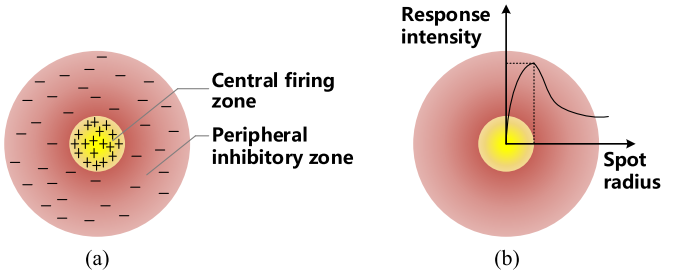


Fig. 6. CRF. (a) Structure of CRF. (b) Spot area response curve.

inhibitory surround with opposite responses [52]. As shown in Fig. 6(a) and (b), the center has a strong but narrow excitatory effect, while the surround has a weaker but broader inhibitory effect, forming an antagonistic mechanism. The spot area response curve shows that the cell response increases with stimulus size until it exceeds the center region, after which inhibition dominates and the response decays.

Inspired by this property, we design an FIM to adjust the RF size for fitting building regions of different scales. The FIM gradually enlarges the RF of deformable convolution [53] to capture multiscale features. Specifically, feature maps are first processed by a 1×1 dynamic convolution for small objects, then split into branches where channel correlations are recalibrated with GAP and 1-D convolution, while other branches use larger kernels (3×3, 5×5, and 7×7) for larger objects. The outputs from multiple scales are concatenated, reduced with a 1×1 convolution, and fused via residual connections to generate the attention-enhanced feature maps.

D. Bidirectional Feature Co-Interaction Module

The BFCIM is designed to alleviate haze-induced feature distortion and semantic ambiguity by dynamically enhancing the interaction between dehazing and segmentation features. In particular, BFCIM enables two-way collaboration between the dehazing and segmentation branches, allowing each task to provide complementary guidance for the other. In the dehazing branch, segmentation features act as queries to refine dehazing features, thereby preserving structural boundaries and generating cleaner restoration results. Conversely, in the segmentation branch, dehazing features are used as queries to guide segmentation, alleviating haze-induced ambiguity and sharpening object boundaries.

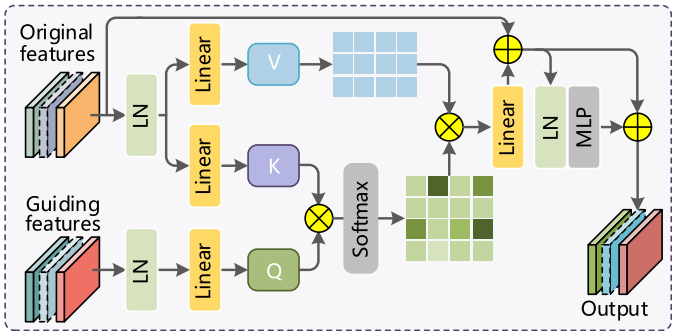


Fig. 7. Structure of BFCIM.

As shown in Fig. 7, the BFCIM enhances feature interaction by taking the query features Q from the guiding branch (segmentation or dehazing) and the key–value features K, V from the current branch. Formally, the module applies multihead cross-attention within local windows to align and fuse the two types of features

$$\text{Attn}(Q,K,V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d}}\right)V \tag{6}$$

where d is the dimension of each head. The attended features are projected back and fused with the original input through residual connections

$$X' = \text{Attn}(Q,K,V) + X \tag{7}$$

where X denotes the original features of the current branch. To further enhance representation, X' is normalized (LN) and refined by a multilayer perceptron (MLP)

$$X_{\text{out}} = \text{MLP}\left(\text{LN}\left(X'\right)\right) + X'. \tag{8}$$

By alternately applying this process in both branches, BFCIM achieves segmentation-guided dehazing and dehazing-guided segmentation. At multiple scales, BFCIM ensures that both low-frequency contextual cues and high-frequency details are collaboratively aligned, thereby enhancing fine-detail restoration as well as semantic boundary localization. This bidirectional design effectively mitigates haze-induced feature distortion and improves both restoration and segmentation performance under challenging conditions.

E. Collaborative Knowledge Transfer

Due to the low contrast, uneven brightness, and cloud occlusion issues in hazy aerial images, the semantic features in the hazy aerial images are corrupted, lacking discriminative information. In contrast, the semantic information in the corresponding clear aerial images is more complete, and the features are more discriminative. Therefore,

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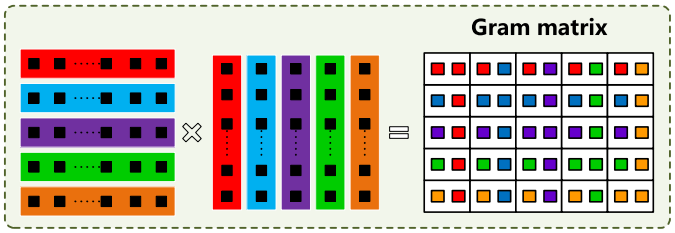


Fig. 8. Calculation process of the Gram matrix.

to improve the segmentation performance of the building segmentation model for hazy aerial images, we propose a CKT framework.

During the training process of D-Net, the pretrained T-Net provides feature information at different stages for D-Net through a Gram matrix-based feature alignment $\mathcal{L}_{Gram}$, logit mimicking $\mathcal{L}_{LM}$, and pseudo-pixel regression $\mathcal{L}_{PR}$ for feature transfer, making the intermediate features, logical feature distribution, and output results of D-Net as close as possible to T-Net. Therefore, CKT is defined as

$$\mathcal{L}_{CKT} = \alpha_1 \mathcal{L}_{Gram} + \alpha_2 \mathcal{L}_{LM} + \alpha_3 \mathcal{L}_{PR} \tag{9}$$

where α_1, α_2, and α_3 denote adjustment factors.

1) Gram Matrix-Based Feature Alignment: To enhance the consistency of deep features between networks, we designed a Gram matrix-based feature alignment module. As shown in Fig. 8, the Gram matrix reflects the correlation between feature points by computing the eccentric covariance matrix of feature vectors. In our framework, this module is used to transfer features from T-Net to D-Net by aligning their Gram matrices. Specifically, we extract feature maps from four different layers of T-Net (the last one after ASPP) and from the corresponding layers of the shared encoder in D-Net. For each pair of layers, we compute the Gram matrices of their features and then measure how close they are.

The alignment is achieved by minimizing the distance between the Gram matrices of T-Net and D-Net, ensuring that the feature representations in D-Net gradually become more consistent with those in T-Net. This process helps the dehazing network inherit both structural and texture knowledge from the T-Net, leading to improved generalization on hazy aerial images. It can be described by the following:

$$\mathcal{L}_{Gram} = \sum_{i=1}^{4} \left\| \mathbb{G}\left(\mathcal{F}_i^t\right) - \mathbb{G}\left(\mathcal{F}_i^d\right) \right\|_2 \tag{10}$$

where $\mathbb{G}(\cdot)$ denotes the Gram matrix. $\mathcal{F}^t$ and $\mathcal{F}^d$ are feature maps in the encoder of T-Net and the shared encoder of D-Net.

2) Logit Mimicking: In this section, logit mimicking is incorporated into CKT to optimize D-Net. As D-Net is a lightweight model, one of the main objectives during the training process of D-Net is to mimic the feature distribution and decision-making capability of the complex model T-Net, thereby rapidly improving its own model performance. If D-Net is directly trained with real labels, the training task becomes excessively difficult for D-Net, resulting in slow optimization speed and poor optimization effectiveness. Therefore, in logit mimicking, D-Net learns soft labels predicted by T-Net, thereby reducing the learning difficulty during the training process of D-Net. This approach allows D-Net to better mimic T-Net and gradually improve the model's robustness and generalization performance.

Specifically, CKT uses the class probability map P^T predicted by T-Net as soft labels, and the class probability map generated by D-Net as P^D. The Kullback–Leibler (KL) divergence is employed to compare the probability distribution similarity between P^T and P^D, achieving logical imitation of D-Net toward T-Net. The calculation process of logit mimicking is illustrated as follows:

$$\mathcal{L}_{LM} = \frac{1}{W \times H} \sum_{i \in \Phi} \mathbb{KL}\left(p_i^D, p_i^T\right) \tag{11}$$

where p_i^D represents the class probability of the ith pixel predicted by D-Net, p_i^T represents the class probability of the ith pixel predicted by T-Net, and $\mathbb{KL}(\cdot)$ represents KL divergence. W and H are the width and height of the image, respectively, and $\Phi = \{1, 2, \ldots, W \times H\}$ is the pixel in the image.

3) Pseudo-Pixel Regression: The aforementioned logit mimicking treats semantic segmentation as a classification problem, utilizing knowledge distillation to optimize the class probability of each pixel generated by D-Net. The pseudo-pixel regression, on the other hand, transforms the semantic segmentation problem into a regression task. During the training process of D-Net, the segmentation result map predicted by T-Net is introduced as an additional regression target, aiming to make the segmentation result of hazy aerial images as close as possible to the segmentation result of clear aerial images, as illustrated in the following equation:

$$\hat{y}_S^D \approx \hat{y}_S^T. \tag{12}$$

To enhance the structural integrity of the segmentation results, the SSIM loss is employed to measure the SSIM between the segmentation results of hazy aerial images and the segmentation results of clear aerial images

$$\mathcal{L}_{PR} = L_{SSIM}\left(\hat{y}_S^D, \hat{y}_S^T\right). \tag{13}$$

F. Joint Optimization Strategy

The proposed CDKL is optimized by a joint optimization strategy under a unified framework to improve the segmentation accuracy of hazy aerial images. The joint optimization strategy is defined as

$$\mathcal{L}_J = \mathcal{L}_{Dehaze} + \mathcal{L}_{seg} + \mathcal{L}_{CKT} \tag{14}$$

where $\mathcal{L}_J$ is the overall loss, which consists of dehazing loss $\mathcal{L}_{Dehaze}$, segmentation loss $\mathcal{L}_{seg}$, and CKT $\mathcal{L}_{CKT}$. All proposed modules in D-Net are trained together.

IV. Experiments

A. Datasets and Metrics

To synthesize hazy aerial image datasets, we add nonuniform haze with different concentrations on the Inria Aerial Image Labeling Dataset (IAILD) [54]. The original IAILD dataset is a building segmentation dataset, which covers an

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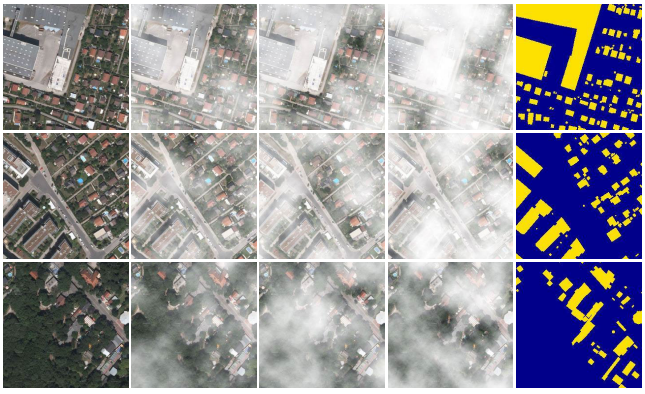


Fig. 9. Example images on the IAILD-T, IAILD-M, and IAILD-D datasets. (a) Clear aerial images. (b) Thin haze. (c) Moderate haze. (d) Dense haze. (e) Labels.

area of 810 km^2 and has a ground resolution of 0.3 m. Due to its ultrahigh ground resolution, the aerial images in the IAILD have complex backgrounds, severe shadow occlusion, and large-scale differences of targets, making the segmentation task itself highly challenging. Our synthetic datasets are based on the IAILD dataset with the addition of three different densities (thin, moderate, and dense) of haze called IAILD-T, IAILD-M, and IAILD-D. Each of the three synthetic datasets contains 14 580 hazy aerial images, in which 14 380 are used as a training set, and the remaining 200 are used as a testing set. The testing sets are called T200, M200, and D200. As shown in Fig. 9, under the influence of uneven haze, aerial images in the hazy IAILD dataset experience reduced contrast, blurred scenes, and loss of detail. These issues degrade the semantic information and significantly increase the complexity of segmentation tasks. In addition, we also evaluate the performance of our model using a real-world hazy UAV dataset, comprising 384 image pairs for training and 50 for testing. The images mainly show low-altitude residential and rural scenes under thin-to-moderate, nonuniform haze, where fine structures are blurred but still recognizable, providing realistic conditions for testing generalization.

We use four indicators to assess our CDKL, including pixel accuracy (PA), mean PA (mPA), mIoU, and frequency-weighted IoU (FWIoU) [55], [56].

B. Implementation Details

The training process of the CDKL model is divided into two stages: In the first stage, the T-Net is trained using the clear aerial images. The input image size is 256×256, the batch size is set to 4, and the total number of training epochs is 61, with a learning rate of 0.00001. In the second stage, with the parameters of the T-Net fixed, the D-Net is trained separately using the hazy aerial images from the IAILD-T, IAILD-M, and IAILD-D datasets. The input image size remains 256×256, and the batch size is set to 10, with a total of 40 training epochs. In both stages, Adam is adopted as the optimizer, with $\beta_1 = 0.9$ and $\beta_2 = 0.999$.

C. Comparison With the State-of-the-Arts

To demonstrate the superiority of our CDKL, we construct 21 "first-enhance-then-segment" compared methods. The dehazing methods include: FCTF-Net [57], DCIL [12] and Trinity-Net [28]. The segmentation methods include: DeepLabv3+ [13], MANet [32], UNetFormer [58], CMTFNet [59], CGGLNet [60], CIGFormer [61], and UrbanSSF [62].

1) Quantitative Evaluation: Table I reports quantitative comparisons on thin, moderate, and dense hazy aerial images (T200/M200/D200), using PA, mPA, mIoU, and FWIoU to evaluate semantic segmentation performance. As shown in Table I, the performance of the same segmentation algorithm varies substantially when paired with different dehazing methods. For instance, UNetFormer [58] achieves its best results when combined with DCIL [12]. Moreover, even with the same dehazing outputs, different segmentation networks exhibit different recognition capabilities, leading to noticeable performance discrepancies. These observations indicate that the "dehazing + segmentation" pipelines are unstable and jointly constrained by both the dehazing quality and the downstream segmentation model. In contrast, CDKL achieves the best performance across all metrics on T200/M200/D200 in Table I, demonstrating the effectiveness of decoupling hazy aerial image segmentation into dehazing and target segmentation within a unified framework.

Furthermore, Table I shows a consistent degradation trend as haze density increases from T200 to M200 and D200: the segmentation accuracy of the compared methods generally declines, suggesting that heavier haze causes more severe spectral and structural degradation and thus has a stronger negative impact on segmentation models. Meanwhile, the improvements brought by conventional "dehazing + segmentation" combinations remain limited and cannot surpass the proposed method. This implies that useful information may be lost and errors may accumulate during the interfacing stage between the dehazing module and the segmentation network, which ultimately degrades the final segmentation performance.

In addition to the "dehazing + segmentation" pipelines, we also compare with state-of-the-art segmentation networks trained directly on hazy aerial images. As summarized in Table I, these methods (e.g., DeepLabv3+, MANet, UNetFormer, CMTFNet, and CGGLNet) yield competitive results under thin haze, but their performance drops markedly as haze concentration increases. For example, on D200, UNetFormer and CGGLNet show a clear reduction in mIoU compared with T200. This demonstrates that training segmentation networks directly on hazy images cannot fully overcome the severe degradation introduced by haze. By contrast, CDKL consistently outperforms all baselines across PA, mPA, mIoU, and FWIoU on T200, M200, and D200 (Table I), validating that explicitly decoupling dehazing and segmentation within a unified framework mitigates error accumulation and yields more robust representations under varying haze densities.

2) Qualitative Evaluation: Figs. 10–12 show the building segmentation results of different models on aerial images of thin haze, moderate haze, and dense haze, respectively. The areas where the segmentation performance is significantly improved are marked with purple boxes. As shown

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TABLE I
QUANTITATIVE EVALUATIONS OF SEMANTIC SEGMENTATION RESULTS ON DATASETS. THE BEST PERFORMANCE IS IN BOLD

Method		T200				M200				D200			
		PA	mPA	mIoU	FWIoU	PA	mPA	mIoU	FWIoU	PA	mPA	mIoU	FWIoU
—	DeepLabv3+	89.07	78.33	65.85	82.15	88.16	71.73	61.49	80.45	87.11	60.50	53.34	77.81
	MANet	86.85	80.45	63.76	79.69	85.75	77.09	61.13	78.21	84.63	69.31	56.55	76.39
	UNetFormer	94.10	89.62	79.32	89.63	93.95	89.60	78.98	89.41	93.05	88.78	76.76	88.08
	CMTFNet	89.01	76.01	64.73	81.87	88.67	75.38	63.97	81.41	88.39	73.86	62.87	80.94
	CGGLNet	94.92	90.18	81.47	90.87	94.78	89.89	81.03	90.64	94.24	88.66	79.37	89.78
	CIGFormer	94.40	86.10	79.88	89.67	92.79	80.51	74.11	86.80	90.10	70.77	64.26	82.06
	UrbanSSF	93.09	87.51	77.47	87.87	92.02	84.01	74.09	86.10	89.45	74.06	65.31	81.77
FCTF-Net	DeepLabv3+	91.85	79.85	68.25	86.47	91.54	79.22	67.44	86.05	91.10	77.32	65.86	85.38
	MANet	90.79	77.03	65.23	84.99	90.71	76.12	64.70	84.83	90.49	73.38	63.06	84.35
	UNetFormer	92.89	78.10	70.43	87.56	92.48	77.06	69.17	86.89	91.71	73.62	65.45	85.82
	CMTFNet	92.57	76.04	68.12	86.96	92.31	75.23	67.72	86.51	91.76	72.72	65.41	85.58
	CGGLNet	93.37	79.21	71.84	88.39	93.18	78.93	71.39	88.06	92.48	75.80	68.26	86.83
	CIGFormer	94.02	89.81	82.70	90.60	93.71	88.93	81.63	90.04	92.98	88.64	80.41	88.82
	UrbanSSF	94.24	88.67	80.37	89.64	93.95	87.77	79.42	89.14	92.99	85.23	76.46	87.55
DCIL	DeepLabv3+	92.29	82.57	70.18	87.20	92.06	82.72	69.81	86.93	91.77	82.52	69.19	86.55
	MANet	91.09	79.65	66.85	85.53	91.05	79.50	66.71	85.47	90.85	78.43	65.93	85.16
	UNetFormer	93.69	82.02	73.64	89.03	93.71	82.26	74.01	89.05	93.36	80.85	72.13	88.46
	CMTFNet	93.62	80.75	72.86	88.86	93.67	81.45	73.73	88.97	93.51	81.09	72.84	88.72
	CGGLNet	94.26	83.15	75.38	89.95	94.38	83.88	76.06	90.15	94.01	82.33	74.98	89.52
	CIGFormer	94.44	90.00	81.15	91.35	94.32	90.29	80.74	91.14	93.94	88.85	80.49	90.47
	UrbanSSF	94.58	90.40	81.66	90.26	94.44	90.19	81.28	90.04	93.85	88.83	79.53	89.07
Trinity-Net	DeepLabv3+	91.53	77.61	67.68	85.87	91.30	76.30	66.92	85.44	90.71	73.90	63.62	84.47
	MANet	90.49	76.82	64.98	84.46	90.36	75.51	64.14	84.13	89.95	72.63	61.37	83.36
	UNetFormer	93.14	79.06	71.77	88.01	92.61	77.47	69.37	87.11	91.48	72.96	65.10	85.14
	CMTFNet	92.94	77.48	69.74	87.64	92.65	76.41	68.60	87.12	91.87	73.81	65.81	85.85
	CGGLNet	93.45	79.35	72.06	88.54	93.24	77.85	70.66	88.07	92.01	74.00	65.55	86.06
	CIGFormer	93.43	88.44	80.76	89.56	93.13	87.32	80.68	89.02	92.21	84.51	79.57	88.42
	UrbanSSF	92.89	82.12	75.02	87.11	92.30	79.73	72.81	86.06	90.94	74.87	67.95	83.68
CDKL (Ours)		**95.72**	**91.26**	**83.88**	**92.17**	**95.69**	**91.01**	**83.72**	**92.10**	**95.42**	**90.38**	**82.83**	**91.65**

in Fig. 10(d), the segmentation results of MANet [32] suffer from under-segmentation due to the interference of haze, making it difficult to generate accurate boundary contours of buildings. Besides, the segmentation results of UNetFormer [58] suffer from a serious over-segmentation phenomenon as shown in Fig. 10(j). As haze concentration increases, its impact on the feature distribution in aerial images becomes more pronounced, making it increasingly difficult for dehazing algorithms to recover the true feature distribution. This results in a steady decline in semantic segmentation performance.

In contrast, the proposed CDKL framework can effectively capture and learn the intrinsic feature associations between hazy and clear aerial images within a unified framework using the CKT system. This system not only facilitates seamless knowledge transfer across different stages of the dehazing process but also ensures that the learned features are well-aligned with both the hazy and clear data distributions. As a result, the framework not only enhances the image restoration quality by recovering more accurate visual details but also significantly boosts semantic segmentation performance. By leveraging both low-level image features and high-level semantic information, the CDKL framework addresses the limitations of conventional dehazing methods, offering a robust solution for processing complex aerial images with varying levels of haze.

D. Hyperparameter Analysis

To determine the optimal balance of the loss components in our CKT framework, defined as (9), we conduct a sensitivity analysis. We vary the weight for each of the three losses individually on the T200 dataset, keeping the other two weights fixed at their determined optimal values. The results are presented in Fig. 13.

1) Analysis of $\mathcal{L}_{Gram}$ Weight α_1: We first analyze the effect of α_1, which controls the feature alignment. The results in Fig. 13(a) show that the model is sensitive to this weight. A value of 1×10^{-8} achieved the peak performance (83.88 % mIoU). Both smaller (1×10^{-9}) and larger (1×10^{-7}) weights led to a degradation in performance, indicating that a precise, small weight is necessary for effective feature transfer.

2) Analysis of $\mathcal{L}_{LM}$ Weight α_2: As shown in Fig. 13(b), we tested α_2 at values of 0.001, 0.01, and 0.1. The model achieved the best results at $\alpha_2 = 0.01$ (83.88% mIoU). Increasing the weight to 0.1 caused a significant drop in performance (81.89% mIoU), suggesting that excessive logit mimicking can harm the student model's optimization.

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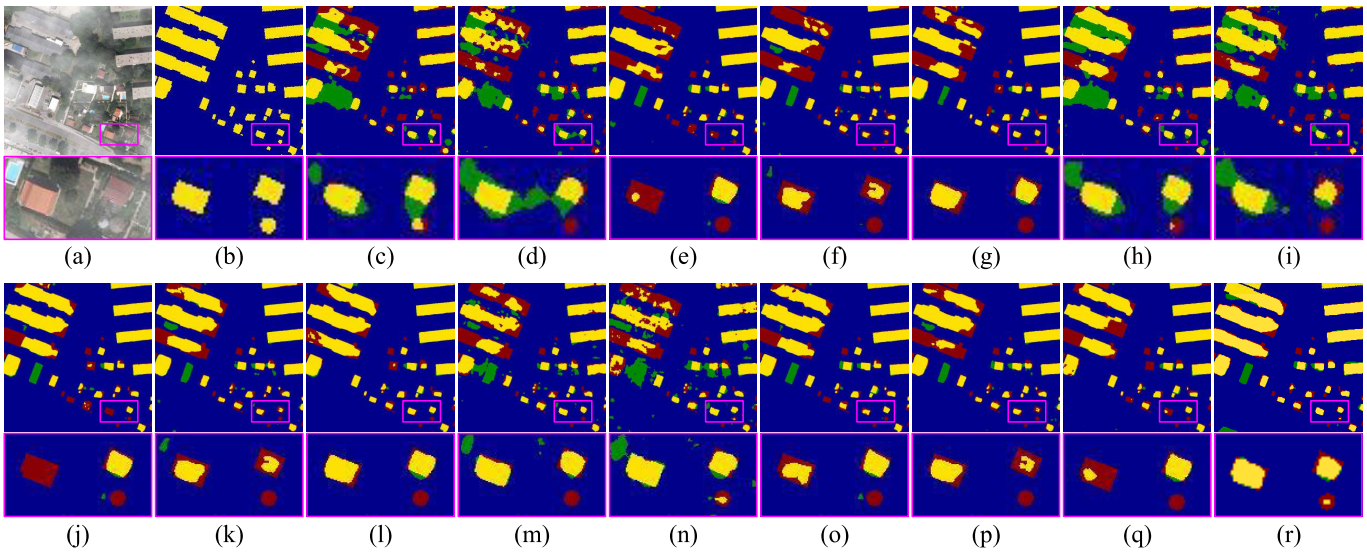


Fig. 10. Segmentation results of comparative methods on T200. The areas with significantly improved segmentation performance are highlighted with purple boxes. Different colors are used for a better view, that is, yellow for true positives, blue for true negatives, green for false positives, and red for false negatives. (a) Hazy aerial images. (b) Ground truths. (c) FCTF-Net [57] + DeepLabv3+ [13]. (d) FCTF-Net [57] + MANet [32]. (e) FCTF-Net [57] + UNetFormer [58]. (f) FCTF-Net [57] + CMTFNet [59]. (g) FCTF-net [57] + CGGLNet [60]. (h) DCIL [12] + DeepLabv3+ [13]. (i) DCIL [12] + MANet [32]. (j) DCIL [12] + UNetFormer [58]. (k) DCIL [12] + CMTFNet [59]. (l) DCIL [12] + CGGLNet [60]. (m) Trinity-net [28] + DeepLabv3+ [13]. (n) Trinity-net [28] + MANet [32]. (o) Trinity-net [28] + UNetFormer [58]. (p) Trinity-net [28] + CMTFNet [59]. (q) Trinity-net [28] + CGGLNet [60]. (r) CDKL (ours).

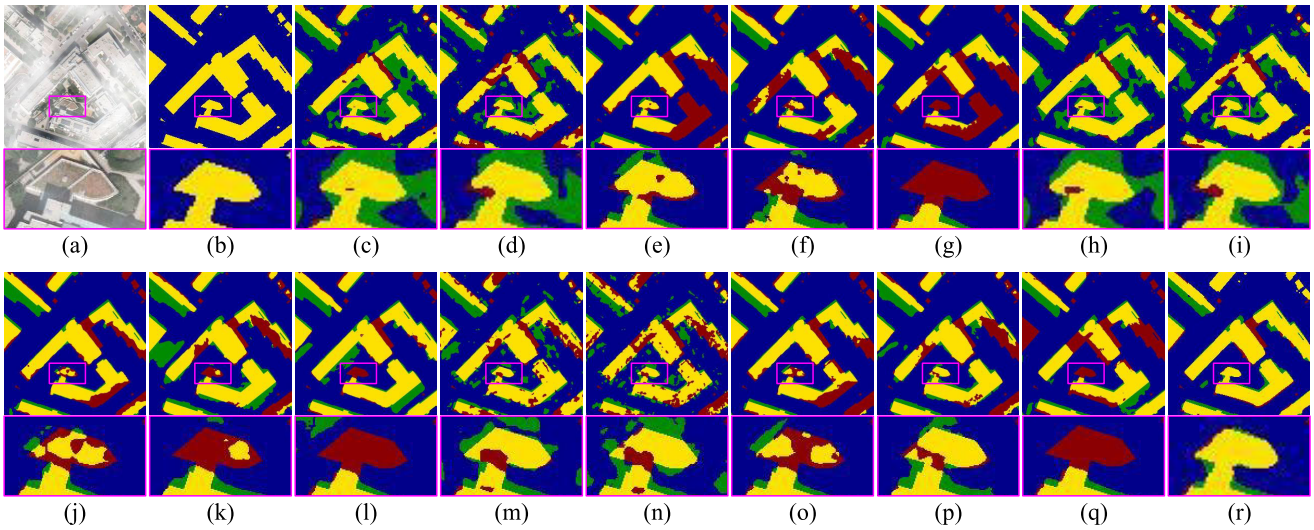


Fig. 11. Segmentation results of comparative methods on M200. The areas with significantly improved segmentation performance are highlighted with purple boxes. Different colors are used for a better view, that is, yellow for true positives, blue for true negatives, green for false positives, and red for false negatives. (a) Hazy aerial images. (b) Ground truths. (c) FCTF-net [57] + DeepLabv3+ [13]. (d) FCTF-Net [57] + MANet [32]. (e) FCTF-net [57] + UNetFormer [58]. (f) FCTF-net [57] + CMTFNet [59]. (g) FCTF-net [57] + CGGLNet [60]. (h) DCIL [12] + DeepLabv3+ [13]. (i) DCIL [12] + MANet [32]. (j) DCIL [12] + UNetFormer [58]. (k) DCIL [12] + CMTFNet [59]. (l) DCIL [12] + CGGLNet [60]. (m) Trinity-net [28] + DeepLabv3+ [13]. (n) Trinity-net [28] + MANet [32]. (o) Trinity-Net [28] + UNetFormer [58]. (p) Trinity-net [28] + CMTFNet [59]. (q) Trinity-net [28] + CGGLNet [60]. (r) CDKL (ours).

3) Analysis of $\mathcal{L}_{PR}$ Weight α_3: Finally, we analyzed α_3, which controls the regression to the teacher's pseudo-labels. The results in Fig. 13(c) indicate that $\alpha_3 = 0.1$ provides the optimal balance, yielding 83.88% mIoU. Both lower (0.01) and higher (1.0) weights lead to slightly lower but still robust performance.

Based on this comprehensive analysis, we select the optimal hyperparameter combination $\alpha_1 = 1 \times 10^{-8}$, $\alpha_2 = 0.01$, and $\alpha_3 = 0.1$ for all experiments presented in this article.

E. Model Complexity Analysis

We comprehensively analyze the model complexity using three metrics: mIoU, number of parameters, and average runtimes on the T200 dataset. All algorithms are conducted on the same equipment. As shown in Table II, the "first-enhance-then-segment" combined algorithms usually have a large number of parameters and long average runtimes but bad performance. On the contrary, the proposed CDKL has the highest mIoU value with the shortest average runtime compared to the other models.

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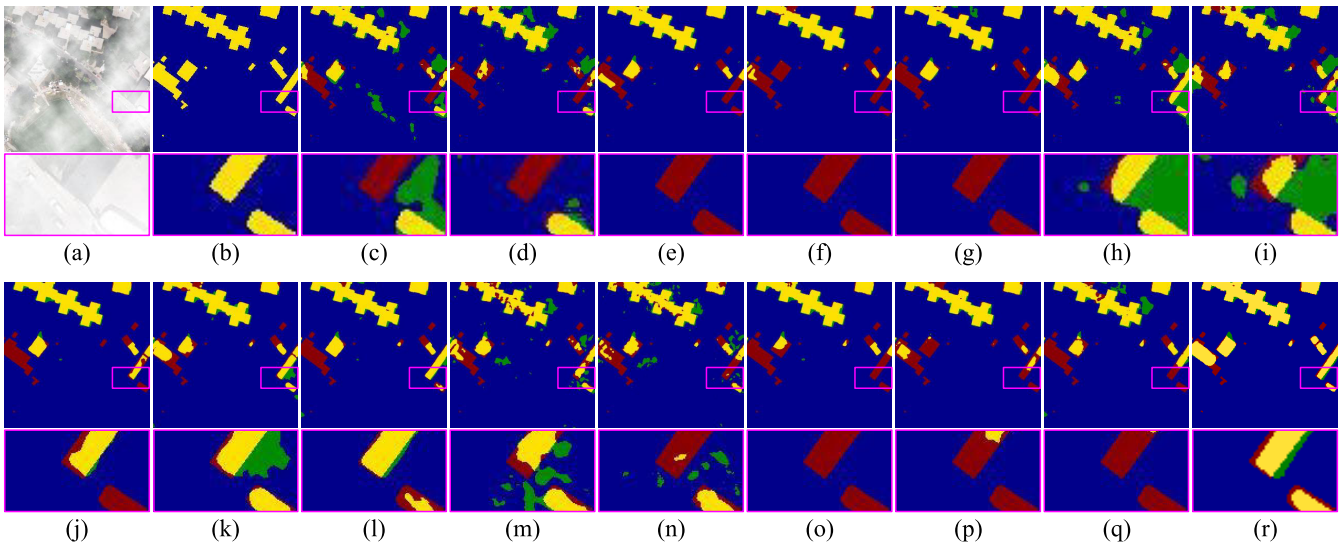


Fig. 12. Segmentation results of comparative methods on D200. The areas with significantly improved segmentation performance are highlighted with purple boxes. Different colors are used for a better view, that is, yellow for true positives, blue for true negatives, green for false positives, and red for false negatives. (a) Hazy aerial images. (b) Ground truths. (c) FCTF-net [57] + DeepLabv3+ [13]. (d) FCTF-net [57] + MANet [32]. (e) FCTF-net [57] + UNetFormer [58]. (f) FCTF-NET [57] + CMTFNet [59]. (g) FCTF-net [57] + CGGLNet [60]. (h) DCIL [12] + DeepLabv3+ [13]. (i) DCIL [12] + MANet [32]. (j) DCIL [12] + UNetFormer [58]. (k) DCIL [12] + CMTFNet [59]. (l) DCIL [12] + CGGLNet [60]. (m) Trinity-net [28] + DeepLabv3+ [13]. (n) Trinity-net [28] + MANet [32]. (o) Trinity-net [28] + UNetFormer [58]. (p) Trinity-net [28] + CMTFNet [59]. (q) Trinity-net [28] + CGGLNet [60]. (r) CDKL (ours).

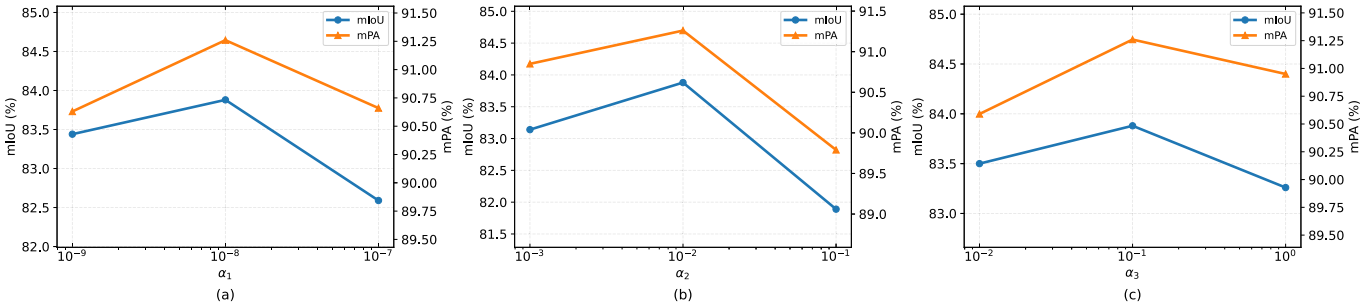


Fig. 13. Sensitivity analysis of CKT loss hyperparameters. (a) $\mathcal{L}_{Gram}$ weight α_1. (b) $\mathcal{L}_{LM}$ weight α_2. (c) $\mathcal{L}_{PR}$ weight α_3.

Therefore, considering the average runtime, number of parameters, and mIoU comprehensively, the proposed CDKL has relatively low model complexity. When processing hazy aerial images, it can maintain good semantic segmentation performance while reducing model complexity.

F. Ablation Analysis

1) Effectiveness of the FIM: We verify the effectiveness of the FIM by removing it from the proposed CDKL. The model without the FIM is denoted as "w/o-FIM." Table III shows the quantitative analysis results of CDKL and w/o-FIM on nonuniform thin hazy aerial images. It can be seen that CDKL achieves higher PA, mPA, mIoU, and FWIoU values compared to w/o-FIM, especially the mPA value, where CDKL outperforms w/o-FIM by 8.47%. Fig. 14 shows the visual comparison of CDKL and w/o-FIM on nonuniform thin hazy aerial images. As shown in the second row of Fig. 14, the FIM module can effectively assist the model in accurately capturing small targets in hazy aerial images, thereby improving the accuracy of the segmentation model.

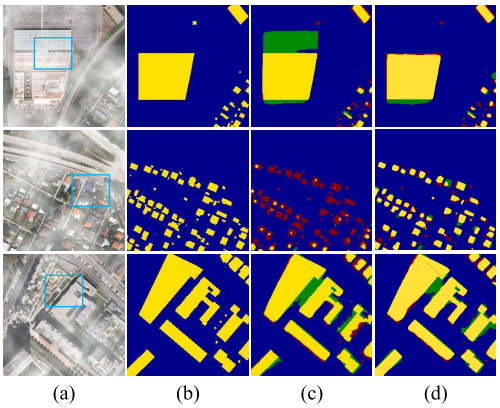


Fig. 14. Ablation study of the FIM on T200. Different colors are used for a better view, that is, yellow for true positives, blue for true negatives, green for false positives, and red for false negatives. (a) Hazy aerial images. (b) Ground truths. (c) w/o FIM. (d) CDKL.

2) Effectiveness of BFCIM: Table III presents the ablation results on the T200 dataset. When BFCIM is removed

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TABLE II

MODEL COMPLEXITY ANALYSIS ON THE T200 DATASET. THE AVERAGE MIOU (%) ↑/NUMBERS OF PARAMETERS (M) ↓/AVERAGE RUNTIMES (MS) ↓ ON DIFFERENT MODELS, ↑ DENOTES THE HIGHER, THE BETTER. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Dehazing	Segmentation	mIoU / Params / Runtimes
FCTF-Net [58]	DeepLabv3+ [13]	68.25 / 39.92 / 52.74
	MANet [32]	65.23 / 36.02 / 49.47
	UNetFormer [58]	70.43 / **11.85** / 48.92
	CMTFNet [59]	68.12 / 30.23 / 79.95
	CGGLNet [60]	71.84 / 48.72 / 86.84
	CIGFormer [61]	82.70 / 52.36 / 50.14
	UrbanSSF [62]	80.37 / 16.36 / 46.23
DCIL [12]	DeepLabv3+ [13]	70.18 / 66.27 / 73.16
	MANet [32]	66.85 / 62.37 / 69.90
	UNetFormer [58]	73.64 / 38.19 / 69.35
	CMTFNet [59]	72.86 / 56.58 / 100.37
	CGGLNet [60]	75.38 / 75.07 / 107.27
	CIGFormer [61]	81.15 / 74.81 / 70.57
	UrbanSSF [62]	81.66 / 38.81 / 66.69
Trinity-Net [28]	DeepLabv3+ [13]	67.68 / 59.90 / 192.01
	MANet [32]	64.98 / 56.01 / 188.74
	UNetFormer [58]	71.77 / 31.83 / 188.20
	CMTFNet [59]	69.74 / 50.22 / 219.22
	CGGLNet [60]	72.06 / 68.71 / 226.12
	CIGFormer [61]	80.76 / 68.45 / 189.41
	UrbanSSF [62]	75.02 / 32.45 / 185.50
CDKL (Ours)		**83.88** / 46.51 / **29.77**

TABLE III

ABLATION STUDY OF THE FIM AND BFCIM ON THE T200 DATASET

	PA (%)	mPA (%)	mIoU (%)	FWIoU (%)
w/o-FIM	94.42	82.79	74.97	90.02
w/o-BFCIM	94.85	86.02	77.32	90.86
CDKL	95.72	91.26	83.88	92.17

(w/o-BFCIM), the performance drops noticeably compared with the full CDKL model, with mPA decreasing from 91.26% to 86.02% and mIoU from 83.88% to 77.32%. This indicates that BFCIM plays a crucial role in enhancing segmentation under hazy conditions. By enabling bidirectional co-interaction between dehazing and segmentation features, BFCIM alleviates haze-induced feature distortion and semantic ambiguity, leading to more accurate boundary delineation and improved recognition of building regions. The consistent improvements across all metrics (PA, mPA, mIoU, and FWIoU) demonstrate that BFCIM effectively complements FIM and significantly contributes to the overall robustness and accuracy of the proposed CDKL framework.

3) Effectiveness of CKT: To verify the effectiveness of each component in the CKT system, we design the following three ablation models: 1) CDKL-I model without $\mathcal{L}_{Gram}$; 2) CDKL-II model without $\mathcal{L}_{LM}$; and 3) CDKL-III model without $\mathcal{L}_{PR}$.

The quantitative results of the ablation experiments on the T200 dataset are summarized in Table IV. Compared with the complete CDKL model, the CDKL-I model without $\mathcal{L}_{Gram}$ sees a decrease of 1.25%, 5.83%, 7.76%, and 1.88% in PA, mPA, mIoU, and FWIoU, respectively. This indicates that the Gram matrix-based feature alignment module can effectively improve the segmentation performance through intermediate feature alignment.

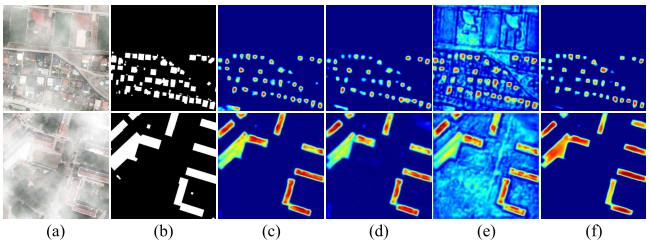


Fig. 15. Visualization of saliency maps in the ablation study of CKT on M200 and D200. (a) Hazy aerial images. (b) Ground truths. (c) CDKL-I. (d) CDKL-II. (e) CDKL-III. (f) CDKL (ours).

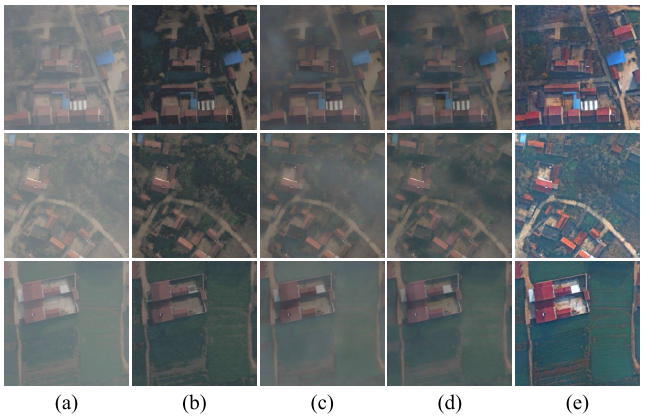


Fig. 16. Dehazing results on the UAV dataset. (a) Hazy UAV images. (b) FCTF-Net [57]. (c) DCIL [12]. (d) Trinity-net [28]. (e) CDKL (ours).

Furthermore, when $\mathcal{L}_{LM}$ is removed from CDKL, the mPA value of CDKL-II decreases by 6.03%, and the mIoU value decreases by 7.61% compared to the complete CDKL model. This shows that by using the logic imitation module to guide D-Net with the soft labels predicted by T-Net, the training difficulty of D-Net can be effectively reduced, thereby improving model performance.

Finally, when $\mathcal{L}_{PR}$ is removed during the training of CDKL, CDKL-III also exhibits a decrease in all performance metrics. The pseudo-pixel regression module can transform the semantic segmentation problem into a regression task, making the model's semantic segmentation results for hazy aerial images more similar to those for clear aerial images.

In addition to the quantitative analysis, we further provide saliency map visualizations to qualitatively validate the motivation of this article. As shown in Fig. 15, CDKL produces sharper and more complete saliency regions, while the ablation models exhibit incomplete or noisy activations. These results demonstrate that the proposed components contribute complementary benefits and jointly enable CKT to achieve superior segmentation under haze conditions.

G. Performance on Real-World Hazy Images

1) Dehazing Comparison: Fig. 16 illustrates the dehazing performance of various dehazing models compared to our

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TABLE IV
ABLATION EXPERIMENTS OF CKT ON T200

Models	$\mathcal{L}_{Gram}$	$\mathcal{L}_{LM}$	$\mathcal{L}_{PR}$	PA (%)	mPA (%)	mIoU (%)	FWIoU (%)
CDKL-I	×	√	√	94.47	85.43	76.12	90.29
CDKL-II	√	×	√	94.56	85.23	76.27	90.40
CDKL-III	√	√	×	94.63	85.03	76.37	90.48
CDKL	√	√	√	95.72	91.26	83.88	92.17

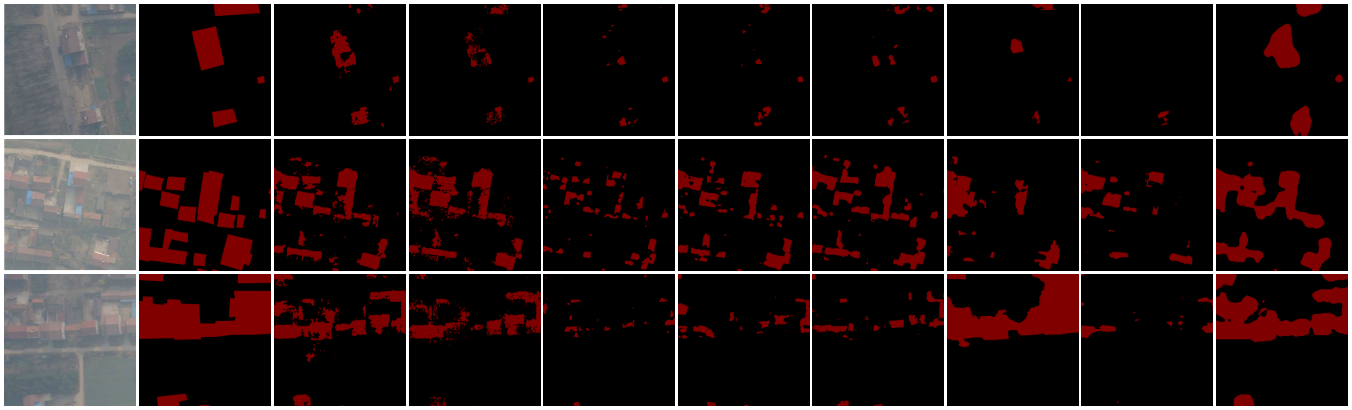


Fig. 17. Segmentation results of comparative methods on real-world hazy images. (a) Hazy UAV images. (b) Ground truths. (c) DeepLabv3+ [13]. (d) MANet [32]. (e) UNetFormer [58]. (f) CMTFNet [59]. (g) CGGLNet [60]. (h) CIGFormer [61]. (i) UrbanSSF [62]. (j) CDKL (ours).

proposed method. As depicted in Fig. 16, the competing dehazing models struggle to remove haze effectively, resulting in images where the building structures remain heavily obscured and lack clarity. The residual haze not only diminishes the overall visibility but also significantly affects the structural details of the buildings. In contrast, our method demonstrates superior dehazing capability, effectively eliminating the haze and restoring the scene with remarkable clarity. The buildings in the dehazed images produced by our approach exhibit enhanced sharpness and saliency, highlighting the effectiveness of our method in preserving structural integrity and improving visual quality.

2) Segmentation Comparison: The poor dehazing capabilities of existing dehazing models make their dehazed results unsuitable for downstream segmentation. Consequently, we utilize the dehazed images generated by our proposed method as input to evaluate the segmentation models. Fig. 17(b) presents the ground truths, which were meticulously annotated by our team. The comparison methods struggle to accurately segment the buildings, primarily due to the challenging nature of the UAV dataset, which includes complex environmental conditions and diverse scene variations. In contrast, our proposed method demonstrates superior performance by accurately detecting and segmenting the building regions. This enhanced accuracy can be attributed to the CKT mechanism incorporated into our model, which effectively bridges the gap between dehazing and segmentation tasks. As a result, our approach not only removes haze but also significantly improves the quality of segmentation outcomes, making it a robust solution for real-world UAV imagery applications.

V. CONCLUSION

Aerial images frequently suffer from haze degradation, which introduces distribution shifts, blurs small or weak structures, and complicates segmentation in complex overlapping scenes. To address these challenges, we proposed the CDKL model, specifically tailored for building segmentation of hazy aerial images. Unlike conventional two-stage pipelines, CDKL integrates dehazing and segmentation into a unified framework. It introduces three key designs: 1) a CKT system, guided by a T-Net trained on clear images, to mitigate domain shifts; 2) a dual-branch decoupled network with an FIM that adapts RFs for buildings of diverse scales and weak signals; and 3) a BFCIM that establishes mutual guidance between dehazing and segmentation, reducing haze-induced semantic ambiguity and improving both detail restoration and boundary localization. Extensive experiments confirm the effectiveness of CDKL, showing clear gains in segmentation accuracy and robustness under nonuniform haze. Nevertheless, we note that the model may not perform optimally in scenarios dominated by thick cloud cover, where ground information is completely obscured. Future work will consider incorporating multimodal signals, such as SAR, to further enhance robustness in such conditions.

REFERENCES

[1] L. Huang, B. Jiang, S. Lv, Y. Liu, and Y. Fu, "Deep-learning-based semantic segmentation of remote sensing images: A survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 8370–8396, 2024.
[2] X. Ma, X. Zhang, M.-O. Pun, and M. Liu, "A multilevel multimodal fusion transformer for remote sensing semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5403215.

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[3] X. Ma, Q. Wu, X. Zhao, X. Zhang, M.-O. Pun, and B. Huang, "SAM-assisted remote sensing imagery semantic segmentation with object and boundary constraints," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5636916.

[4] G. Y. Lee, J. Chen, T. Dam, M. M. Ferdaus, D. P. Poenar, and V. N. Duong, "Dehazing remote sensing and UAV imagery: A review of deep learning, prior-based, and hybrid approaches," 2024, *arXiv:2405.07520*.

[5] C.-H. Lin, K.-H. Lai, Z.-B. Chen, and J.-Y. Chen, "Patch-based information reconstruction of cloud-contaminated multitemporal images," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 163–174, Jan. 2014.

[6] B. Sui, Y. Cao, X. Bai, S. Zhang, and R. Wu, "BIBED-seg: Block-in-block edge detection network for guiding semantic segmentation task of high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1531–1549, 2023.

[7] J. Nie, L. Huang, C. Zheng, X. Lv, and R. Wang, "Cross-scale graph interaction network for semantic segmentation of remote sensing images," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 19, no. 6, pp. 1–18, Nov. 2023.

[8] C. Pan, R. Li, W. Liu, W. Lu, C. Niu, and Q. Bao, "Remote sensing image ship detection based on dynamic adjusting labels strategy," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4702621.

[9] C. Chen, W. Zeng, and X. Zhang, "HFPNet: Super feature aggregation pyramid network for maritime remote sensing small-object detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 5973–5989, 2023.

[10] X. Ma, X. Zhang, and M.-O. Pun, "RS^{3}Mamba: Visual state space model for remote sensing image semantic segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[11] X. Li et al., "A spectral–spatial context-boosted network for semantic segmentation of remote sensing images," *Remote Sens.*, vol. 16, no. 7, p. 1214, Mar. 2024. [Online]. Available: https://www.mdpi.com/2072-4292/16/7/1214

[12] L. Zhang and S. Wang, "Dense haze removal based on dynamic collaborative inference learning for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5631016.

[13] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking Atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[14] L. Xu, D. Zhao, Y. Yan, S. Kwong, J. Chen, and L.-Y. Duan, "IDeRs: Iterative dehazing method for single remote sensing image," *Inf. Sci.*, vol. 489, pp. 50–62, Jul. 2019.

[15] H. Shen, C. Zhang, H. Li, Q. Yuan, and L. Zhang, "A spatial–spectral adaptive haze removal method for visible remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6168–6180, Sep. 2020.

[16] F. A. Dharejo, Y. Zhou, F. Deeba, M. A. Jatoi, Y. Du, and X. Wang, "A remote-sensing image enhancement algorithm based on patch-wise dark channel prior and histogram equalisation with colour correction," *IET Image Process.*, vol. 15, no. 1, pp. 47–56, Jan. 2021.

[17] G. Bi, G. Si, Y. Zhao, B. Qi, and H. Lv, "Haze removal for a single remote sensing image using low-rank and sparse prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5615513.

[18] Y. Han, M. Yin, P. Duan, and P. Ghamisi, "Edge-preserving filtering-based dehazing for remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[19] D. Singh, M. Kaur, M. Y. Jabarulla, V. Kumar, and H.-N. Lee, "Evolving fusion-based visibility restoration model for hazy remote sensing images using dynamic differential evolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 1002214.

[20] L. Jiao, C. Hu, L. Huo, and P. Tang, "Guided-Pix2Pix: End-to-end inference and refinement network for image dehazing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3052–3069, 2021.

[21] Y. Zheng, J. Su, S. Zhang, M. Tao, and L. Wang, "Dehaze-TGGAN: Transformer-guide generative adversarial networks with spatial-spectrum attention for unpaired remote sensing dehazing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5634320.

[22] H. Su, L. Liu, G. Jeon, Z. Wang, T. Guo, and M. Gao, "Remote sensing image dehazing based on dual attention parallelism and frequency domain selection network," *IEEE Trans. Consum. Electron.*, vol. 70, no. 3, pp. 5300–5311, Aug. 2024.

[23] X. Chen, Y. Li, L. Dai, and C. Kong, "Hybrid high-resolution learning for single remote sensing satellite image dehazing," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[24] J. Nie, W. Wei, L. Zhang, J. Yuan, Z. Wang, and H. Li, "Contrastive haze-aware learning for dynamic remote sensing image dehazing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5634311.

[25] X. Chen and Y. Huang, "Memory-oriented unpaired learning for single remote sensing image dehazing," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[26] Y. Zheng, J. Su, S. Zhang, M. Tao, and L. Wang, "Dehaze-AGGAN: Unpaired remote sensing image dehazing using enhanced attention-guide generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5630413.

[27] Y. Wen, T. Gao, Z. Li, J. Zhang, and T. Chen, "Encoder-minimal and decoder-minimal framework for remote sensing image dehazing," in *Proc. ICASSP - IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 36–40.

[28] K. Chi, Y. Yuan, and Q. Wang, "Trinity-Net: Gradient-guided Swin transformer-based remote sensing image dehazing and beyond," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4702914.

[29] N. Wang et al., "Weakly supervised image dehazing via physics-based decomposition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 1, pp. 637–652, Jan. 2026.

[30] Y. Su, N. Wang, Z. Cui, Y. Cai, C. He, and A. Li, "Real scene single image dehazing network with multi-prior guidance and domain transfer," *IEEE Trans. Multimedia*, vol. 27, pp. 5492–5506, 2025.

[31] F. Xu, Y. Shi, W. Yang, G.-S. Xia, and X. X. Zhu, "CloudSeg: A multi-modal learning framework for robust land cover mapping under cloudy conditions," *ISPRS J. Photogramm. Remote Sens.*, vol. 214, pp. 21–32, Aug. 2024.

[32] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607713.

[33] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.

[34] X. Li et al., "A synergistical attention model for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5400916.

[35] J. Jin, W. Zhou, R. Yang, L. Ye, and L. Yu, "Edge detection guide network for semantic segmentation of remote-sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[36] Q. Zhao, J. Liu, Y. Li, and H. Zhang, "Semantic segmentation with attention mechanism for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5403913.

[37] T. Xiao, Y. Liu, Y. Huang, M. Li, and G. Yang, "Enhancing multiscale representations with transformer for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605116.

[38] W. Zhou, X. Fan, W. Yan, S. Shan, Q. Jiang, and J.-N. Hwang, "Graph attention guidance network with knowledge distillation for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4506015.

[39] Y. Wang, G. Sun, and S. Guo, "Target detection method for low-resolution remote sensing image based on ESRGAN and ReDet," *Photonics*, vol. 8, no. 10, p. 431, Oct. 2021.

[40] M. B. Pereira and J. A. D. Santos, "An end-to-end framework for low-resolution remote sensing semantic segmentation," in *Proc. IEEE Latin Amer. GRSS ISPRS Remote Sens. Conf. (LAGIRS)*, Mar. 2020, pp. 6–11.

[41] Q. Zhang, G. Yang, and G. Zhang, "Collaborative network for super-resolution and semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2021.

[42] Y. Wei, Z. Zhang, H. Zheng, R. Hong, Y. Yang, and M. Wang, "SGINet: Toward sufficient interaction between single image deraining and semantic segmentation," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 6202–6210.

[43] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.

[44] Y. Liu, K. Chen, C. Liu, Z. Qin, Z. Luo, and J. Wang, "Structured knowledge distillation for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2604–2613.

[45] Z. Dong, G. Gao, T. Liu, Y. Gu, and X. Zhang, "Distilling segmenters from CNNs and transformers for remote sensing images' semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5613814.

[46] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[47] Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, "Residual dense network for image restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 7, pp. 2480–2495, Jul. 2021.

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[48] Y. Xue et al., "FMTrack: Frequency-aware interaction and multi-expert fusion for RGB-T tracking," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 2, pp. 1–13, Feb. 2026.
[49] Y. Xue et al., "AVLTrack: Dynamic sparse learning for aerial vision-language tracking," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 8, pp. 7554–7567, Aug. 2025.
[50] Y. Xue et al., "Target-distractor aware UAV tracking via global agent," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 10, pp. 16116–16127, Oct. 2025.
[51] X. Qin, Z. Wang, Y. Bai, X. Xie, and H. Jia, "FFA-Net: Feature fusion attention network for single image dehazing," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 11908–11915.
[52] S. W. Kuffler, "Discharge patterns and functional organization of mammalian retina," *J. Neurophysiology*, vol. 16, no. 1, pp. 37–68, Jan. 1953.
[53] X. Zhu, H. Hu, S. Lin, and J. Dai, "Deformable ConvNets v2: More deformable, better results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9308–9316.
[54] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Can semantic labeling methods generalize to any city? The Inria aerial image labeling benchmark," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 3226–3229.
[55] Y. Mo, Y. Wu, X. Yang, F. Liu, and Y. Liao, "Review the state-of-the-art technologies of semantic segmentation based on deep learning," *Neurocomputing*, vol. 493, pp. 626–646, Jul. 2022.
[56] S. Hao, Y. Zhou, and Y. Guo, "A brief survey on semantic segmentation with deep learning," *Neurocomputing*, vol. 406, pp. 302–321, Sep. 2020.
[57] Y. Li and X. Chen, "A coarse-to-fine two-stage attentive network for haze removal of remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 10, pp. 1751–1755, Oct. 2021.
[58] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271622001654
[59] H. Wu, P. Huang, M. Zhang, W. Tang, and X. Yu, "CMTFNet: CNN and multiscale transformer fusion network for remote-sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2004612.
[60] Y. Ni, J. Liu, W. Chi, X. Wang, and D. Li, "CGGLNet: Semantic segmentation network for remote sensing images based on category-guided global–local feature interaction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5615617.
[61] Y. Ni, J. Liu, H. Zhang, W. Chi, and J. Luan, "Category-guided transformer for semantic segmentation of high-resolution remote sensing images," *Remote Sens.*, vol. 17, no. 17, p. 3054, Sep. 2025.
[62] Z. Wang, J. Yi, A. Chen, L. Chen, H. Lin, and K. Xu, "Accurate semantic segmentation of very high-resolution remote sensing images considering feature state sequences: From benchmark datasets to urban applications," *ISPRS J. Photogramm. Remote Sens.*, vol. 220, pp. 824–840, Feb. 2025.



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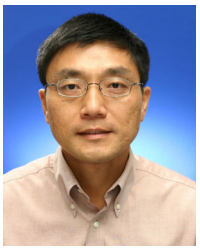
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